

Selection on Observables II: Estimation

PS200C, Spring 2026

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Today: how to actually do the adjustment on the chosen X

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Under CI, these are all estimating the same thing. They differ only in *how they approximate conditioning on X* .

Recap: Subclassification

From SOO I:

- Partition units by X into strata $x_1, x_2, \dots$
- Within each stratum, treated and control are exchangeable (CI).
- Compute DIM within each stratum; weight up to the population:

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Problematic with too many variables, continuous variables: empty and under-populated strata.

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A matched sample is just a flexible subclassification. Same idea; the strata move with the data.

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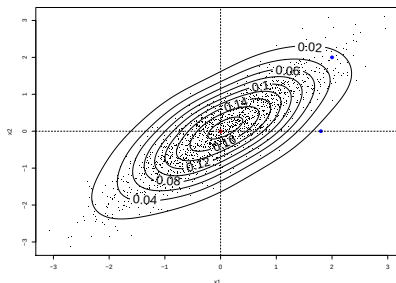
- **Euclidean:** $\|X_i - X_j\|$ — scale-sensitive.
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Problem: unlike true stratification, the treated and control getting compared aren't exactly the same on X_i .

- There is a *discrepancy*, $\|X_i - X_j\|$, between treated and control units in a pair $\rightarrow$ difference in $Y_i(1)$ and $Y_j(0)$. This bias does not vanish at the usual $\sqrt{n}$ rate (Abadie & Imbens, 2006).
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One solution: **bias adjustment** (Abadie & Imbens, 2011).

- Regress $Y \sim X$ among controls; call the fit $\hat{\mu}_0(X)$.
- Use $\hat{\mu}_0(X_i) - \hat{\mu}_0(X_{j(i)})$ to estimate the $Y(0)$ gap implied by the X -discrepancy, and subtract it from each matched pair.

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Instead of matching in high-dimensional X , just a scalar!

Propensity score matching, on Blattman & Annan example

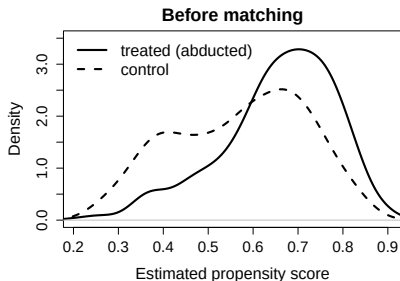
Recipe:

- 1 Fit $\hat{\pi}(X)$ (e.g., logistic regression of D on X).
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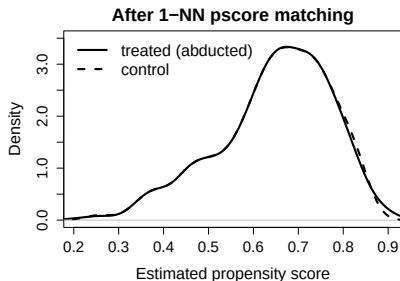
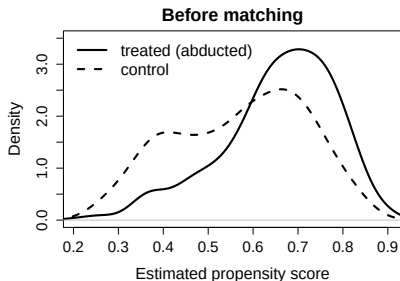
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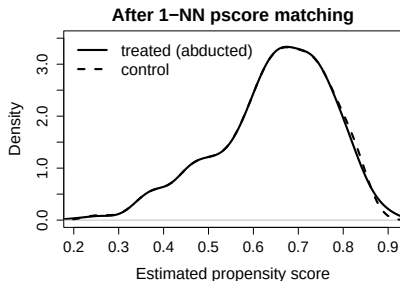
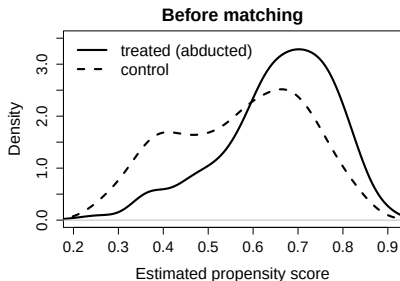
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Caveats: Extreme π may need to be trimmed; depends on how well we estimate $\pi(X)$

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Treated units with low π : upweighted (they're rare).
Control units with high π : upweighted (also rare).

Weights: What we want vs what we have

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Weights can do this...

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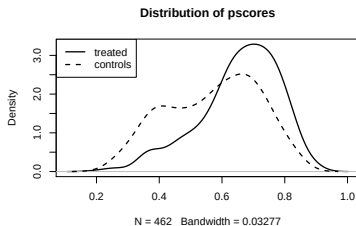
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Drop the $p(D_i)$ in numerator and you'd have the plain IPW above ($1/\hat{\pi}(X_i)$ or $1/(1 - \hat{\pi}(X_i))$), which has higher variance.

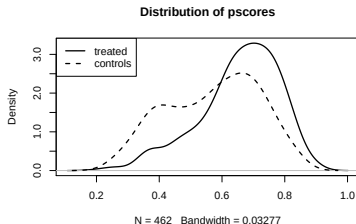
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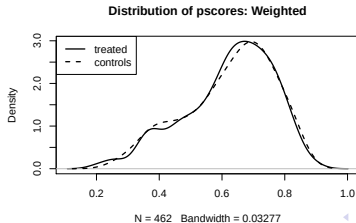


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After IPW, the control distribution is reshaped to match the treated:



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- **Misspecification.** $\hat{\pi}(X)$ wrong $\Rightarrow$ weights wrong $\Rightarrow$ bias.
- Trimming is sometimes used to address extreme weights...changes the estimate, can be arbitrary.

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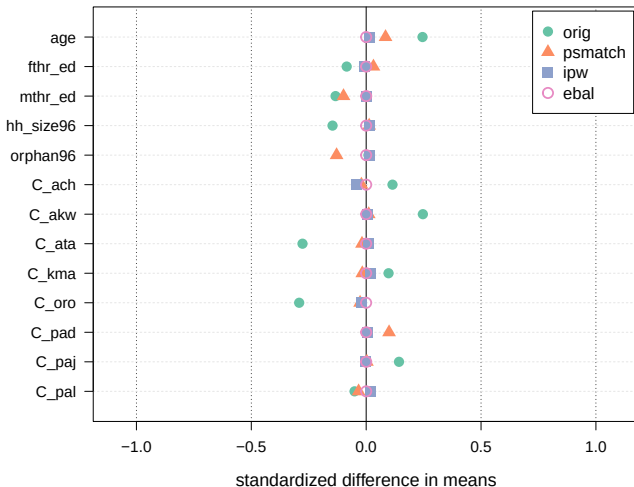
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E.g. maximum entropy balancing (Hainmueller, 2012) minimizes $\sum_i w_i \log(w_i)$ while achieving the target balance conditions.

Balance on Blattman & Annan



Standardized mean differences, 13 covariates. "orig" = no adjustment.

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Tradeoff: you only ensure balance on the things you target, but you get perfect balance on them in the sample, don't need correct pscore.

Regression — start with imputation

What's actually missing? For each unit i with covariates X_i :

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This is **regression imputation**, a.k.a. ***g-computation***. As natural a use of regression as there is.

Linear models $\Rightarrow$ Lin's estimator

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This is **Lin's (2013) estimator** — which you already saw in experiments. The same thing, just now motivated by “impute each unit's missing PO.”

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- Assumes the same slope in X for treated and control.
- Equivalent to fitting a pooled $\hat{\mu}(x)$ and taking $\hat{\tau}$ from the D coefficient.
- When treatment effects vary with X : $\hat{\tau}$ is a weighted average of unit-level effects, with weights that tilt toward units in the region of overlap — not necessarily the ATE or ATT.

What about pooled OLS?

The usual regression $Y = \alpha + \tau D + \beta'X + \varepsilon$ (no interaction):

- Assumes the same slope in X for treated and control.
- Equivalent to fitting a pooled $\hat{\mu}(x)$ and taking $\hat{\tau}$ from the D coefficient.
- When treatment effects vary with X : $\hat{\tau}$ is a weighted average of unit-level effects, with weights that tilt toward units in the region of overlap — not necessarily the ATE or ATT.

Fix: either use the interacted (Lin) version, or commit to the assumption that slopes are equal.

All roads lead to the same place

Under CI, every method on the menu is trying to estimate

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- F is a choice of estimand: $F = p(X)$ gives the ATE; $p(X \mid D=1)$ gives the ATT; $p(X \mid D=0)$ gives the ATC.
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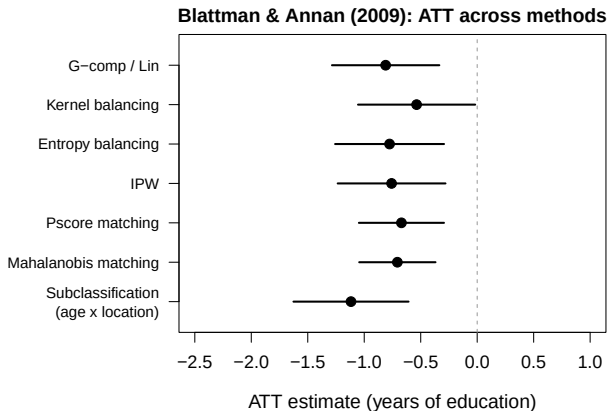
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They achieve this by different strategies:

Method	How it conditions on X
Subclassification	coarse strata of X
Matching	nearest-neighbor “strata” in X
Pscore matching	match on $\hat{\pi}(X)$
IPW	reweight by $1/\hat{\pi}(X)$
Balancing weights	solve for weights balancing X
Regression imputation	model $\mu_d(X)$ and impute

Blattman & Annan: ATT across methods



Covariates: age, fthr_ed, mthr_ed, hh_size96, orphan96, and 8 location dummies.

If everything agrees, great; if not, diagnose.

Beyond: combining pscore and outcome model

So far: estimate either $\hat{\pi}(X)$ (weighting) or $\hat{\mu}_d(X)$ (regression).

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Doubly robust methods combine both:

- **AIPW**: IPW estimator + regression adjustment using $\hat{\mu}_d(X)$.
- **TMLE**: iteratively update $\hat{\mu}_d$ using $\hat{\pi}$.
- **Double ML (DML)**: plug flexible (ML-based) $\hat{\pi}$ and $\hat{\mu}_d$ into AIPW with cross-fitting.

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All still need CI + common support. Flexibility buys you less bias from model form, not from missing confounders.

Takeaways

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- **Regression** is natural when framed as imputation: model $\mu_d(X)$, predict each unit's missing PO (or both), average over intended group.
- **Reasonable practice:** In a given setting, you might take one estimator as central, but good to show range of answers you get under whichever approaches are appropriate.